

## Q1 - 24 January - Shift 1

Let  $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}$ ,  $\vec{v} = 2\hat{i} + \hat{j} - \hat{k}$ ,  $\vec{v} \cdot \vec{w} = 2$  and

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$\vec{v} \times \vec{w} = \vec{u} + \lambda \vec{v}$ . Then  $\vec{u} \cdot \vec{w}$  is equal to

(1) 1

(2)  $\frac{3}{2}$

(3) 2

(4)  $-\frac{2}{3}$

## Q2 - 24 January - Shift 2

Let  $\vec{\alpha} = 4\hat{i} + 3\hat{j} + 5\hat{k}$  and  $\vec{\beta} = \hat{i} + 2\hat{j} - 4\hat{k}$ . Let  $\vec{\beta}_1$  be parallel to  $\vec{\alpha}$  and  $\vec{\beta}_2$  be perpendicular to  $\vec{\alpha}$ . If

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$\vec{\beta} = \vec{\beta}_1 + \vec{\beta}_2$ , then the value of  $5\vec{\beta}_2 \cdot (\hat{i} + \hat{j} + \hat{k})$  is

(1) 6

(2) 11

(3) 7

(4) 9

## Q3 - 24 January - Shift 2

Let  $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}$ ,  $\vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}$ ,  $\vec{a} \cdot \vec{c} = 7$ ,

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$2\vec{b} \cdot \vec{c} + 43 = 0$ ,  $\vec{a} \times \vec{c} = \vec{b} \times \vec{c}$ . Then  $|\vec{a} \cdot \vec{b}|$  is equal to

## Q4 - 25 January - Shift 1

The vector  $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$  is rotated through a right angle, passing through the y-axis in its way and the resulting vector is  $\vec{b}$ . Then the projection of  $3\vec{a} + \sqrt{2}\vec{b}$  on  $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$  is

- (1)  $3\sqrt{2}$  (2) 1  
(3)  $\sqrt{6}$  (4)  $2\sqrt{3}$

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#### Q5 - 25 January - Shift 1

Let  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  be three non zero vectors such that

$\vec{b} \cdot \vec{c} = 0$  and  $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2}$ . If  $\vec{d}$  be a vector

such that  $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$ , then  $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$  is equal to

- (1)  $\frac{3}{4}$  (2)  $\frac{1}{2}$   
(3)  $-\frac{1}{4}$  (4)  $\frac{1}{4}$

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#### Q6 - 25 January - Shift 2

If the four points, whose position vectors are

$3\hat{i} - 4\hat{j} + 2\hat{k}$ ,  $\hat{i} + 2\hat{j} - \hat{k}$ ,  $-2\hat{i} - \hat{j} + 3\hat{k}$  and  $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$  are coplanar, then  $\alpha$  is equal to

- (1)  $\frac{73}{17}$  (2)  $-\frac{107}{17}$   
(3)  $-\frac{73}{17}$  (4)  $\frac{107}{17}$

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#### Q7 - 25 January - Shift 2

Let  $\vec{a} = -\hat{i} - \hat{j} + \hat{k}$ ,  $\vec{a} \cdot \vec{b} = 1$  and  $\vec{a} \times \vec{b} = \hat{i} - \hat{j}$ . Then

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$\vec{a} - 6\vec{b}$  is equal to

(1)  $3(\hat{i} - \hat{j} - \hat{k})$  (2)  $3(\hat{i} + \hat{j} + \hat{k})$

(3)  $3(\hat{i} - \hat{j} + \hat{k})$  (4)  $3(\hat{i} + \hat{j} - \hat{k})$

#### Q8 - 29 January - Shift 1

If the vectors  $\vec{a} = \lambda\hat{i} + \mu\hat{j} + 4\hat{k}$ ,  $\vec{b} = 2\hat{i} + 4\hat{j} - 2\hat{k}$  and  $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$  are coplanar and the projection of  $\vec{a}$  on the vector  $\vec{b}$  is  $\sqrt{54}$  units, then the sum of all possible values of  $\lambda + \mu$  is equal to

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- (1) 0 (2) 6  
(3) 24 (4) 18

#### Q9 - 29 January - Shift 1

Let  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  be three non-zero non-coplanar vectors. Let the position vectors of four points A,

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B, C and D be  $\vec{a} - \vec{b} + \vec{c}$ ,  $\lambda\vec{a} - 3\vec{b} + 4\vec{c}$ ,  $-\vec{a} + 2\vec{b} - 3\vec{c}$  and  $2\vec{a} - 4\vec{b} + 6\vec{c}$  respectively. If  $\overline{AB}$ ,  $\overline{AC}$  and  $\overline{AD}$  are coplanar, then  $\lambda$  is :

#### Q10 - 29 January - Shift 2

Let  $\vec{a} = 4\hat{i} + 3\hat{j}$  and  $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$  and  $\vec{c}$  is a

vector such that  $\vec{c} \cdot (\vec{a} \times \vec{b}) + 25 = 0$ ,  $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$

and projection of  $\vec{c}$  on  $\vec{a}$  is 1, then the projection

of  $\vec{c}$  on  $\vec{b}$  equals :

(1)  $\frac{5}{\sqrt{2}}$

(2)  $\frac{1}{5}$

(3)  $\frac{1}{\sqrt{2}}$

(4)  $\frac{3}{\sqrt{2}}$

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#### Q11 - 29 January - Shift 2

If  $\vec{a} = \hat{i} + 2\hat{k}$ ,  $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ ,  $\vec{c} = 7\hat{i} - 3\hat{k} + 4\hat{k}$ ,

$\vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$  and  $\vec{r} \cdot \vec{a} = 0$  then  $\vec{r} \cdot \vec{c}$  is equal to

(1) 34

(2) 12

(3) 36

(4) 30

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#### Q12 - 30 January - Shift 1

If  $\vec{a}, \vec{b}, \vec{c}$  are three non-zero vectors and  $\hat{n}$  is a unit

vector perpendicular to  $\vec{c}$  such that

$\vec{a} = \alpha \vec{b} - \hat{n}, (\alpha \neq 0)$  and  $\vec{b} \cdot \vec{c} = 12$ , then

$|\vec{c} \times (\vec{a} \times \vec{b})|$  is equal to :

(1) 15

(2) 9

(3) 12

(4) 6

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**Q13 - 30 January - Shift 2**

Let  $\lambda \in \mathbb{R}, \vec{a} = \lambda \hat{i} + 2\hat{j} - 3\hat{k}, \vec{b} = \hat{i} - \lambda \hat{j} + 2\hat{k}$ .

If  $\left( (\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b}) \right) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$ ,

then  $\left| \lambda (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) \right|^2$  is equal to

(1) 140

(2) 132

(3) 144

(4) 136

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**Q14 - 30 January - Shift 2**

Let  $\vec{a}$  and  $\vec{b}$  be two vectors. Let  $|\vec{a}| = 1, |\vec{b}| = 4$  and

$\vec{a} \cdot \vec{b} = 2$ . If  $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ , then the value of

$\vec{b} \cdot \vec{c}$  is

(1) -24

(2) -48

(3) -84

(4) -60

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**Q15 - 31 January - Shift 1**

Let  $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ , and  $\vec{b}$  and  $\vec{c}$  be two nonzero

vectors such that  $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$  and

$\vec{b} \cdot \vec{c} = 0$ . Consider the following two statement:

(A)  $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$  for all  $\lambda \in \mathbb{R}$ .

(B)  $\vec{a}$  and  $\vec{c}$  are always parallel

(1) only (B) is correct

(2) neither (A) nor (B) is correct

(3) only (A) is correct

(4) both (A) and (B) are correct.

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#### Q16 - 31 January - Shift 1

Let  $\vec{a}$  and  $\vec{b}$  be two vector such that  $|\vec{a}| = \sqrt{14}$ ,

$|\vec{b}| = \sqrt{6}$  and  $|\vec{a} \times \vec{b}| = \sqrt{48}$ . Then  $(\vec{a} \cdot \vec{b})^2$  is equal to

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#### Q17 - 31 January - Shift 2

Let :  $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ ,  $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$  and

$\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$  be there vectors. If  $\vec{r}$  is a

vector such that,  $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$  and  $\vec{r} \cdot \vec{a} = 0$ . Then

$25|\vec{r}|^2$  is equal to

(1) 449

(2) 336

(3) 339

(4) 560

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#### Q18 - 31 January - Shift 2

The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points A, B, C is  $(2, a, 4)$ ,  $a \in \mathbb{N}$ . If the volume of the tetrahedron OABC is  $144 \text{ unit}^3$ , then which of the following points is NOT on P?

- (1)  $(2, 2, 4)$
- (2)  $(0, 4, 4)$
- (3)  $(3, 0, 4)$
- (4)  $(0, 6, 3)$

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Q19 - 31 January - Shift 2

Let  $\vec{a}, \vec{b}, \vec{c}$  be three vectors such that

$$|\vec{a}| = \sqrt{31}, 4|\vec{b}| = |\vec{c}| = 2 \text{ and } 2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a}).$$

If the angle between  $\vec{b}$  and  $\vec{c}$  is  $\frac{2\pi}{3}$ , then

$$\left( \frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}} \right)^2 \text{ is equal to } \underline{\hspace{2cm}}.$$

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Q20 - 01 February - Shift 1

Let  $\vec{v} = \alpha \hat{i} + 2\hat{j} - 3\hat{k}$ ,  $\vec{w} = 2\alpha \hat{i} + \hat{j} - \hat{k}$ , and  $\vec{u}$  be a

vector such that  $|\vec{u}| = \alpha > 0$ . If the minimum value

of the scalar triple product  $[\vec{u}\vec{v}\vec{w}]$  is  $-\alpha\sqrt{3401}$ ,

and  $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$  where m and n are coprime natural numbers, then m + n is equal to \_\_\_\_\_.

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Q21 - 01 February - Shift 1

A(2, 6, 2), B(-4, 0,  $\lambda$ ), C(2, 3, -1) and D(4, 5, 0),  
 $|\lambda| \leq 5$  are the vertices of a quadrilateral ABCD. If  
 its area is 18 square units, then  $5 - 6\lambda$  is equal to

\_\_\_\_\_.

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### Q22 - 01 February - Shift 2

Let  $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$  and  $\vec{b} = \hat{i} + 3\hat{j} + 5\hat{k}$  be two vectors. Then which one of the following statements is TRUE ?

1. Projection of  $\vec{a}$  on  $\vec{b}$  is  $\frac{17}{\sqrt{35}}$  and the direction of the projection vector is same as of  $\vec{b}$ .
2. Projection of  $\vec{a}$  on  $\vec{b}$  is  $\frac{17}{\sqrt{35}}$  and the direction of the projection vector is opposite to the direction of  $\vec{b}$ .
3. Projection of  $\vec{a}$  on  $\vec{b}$  is  $\frac{-17}{\sqrt{35}}$  and the direction of the projection vector is same as of  $\vec{b}$ .
4. Projection of  $\vec{a}$  on  $\vec{b}$  is  $\frac{-17}{\sqrt{35}}$  and the direction of the projection vector is opposite to the direction of  $\vec{b}$ .

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### Q23 - 01 February - Shift 2

Let  $\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}$ ,  $\vec{b} = \hat{i} + \hat{k}$  and  
 $\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$  be three given vectors. If  $\vec{r}$  is a  
 vector such that  $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$  and  $\vec{r} \cdot \vec{b} = 0$ , then

$|\vec{r}|$  is equal to :

(1)  $\frac{11}{7}\sqrt{2}$

(2)  $\frac{11}{7}$

(3)  $\frac{11}{5}\sqrt{2}$

(4)  $\frac{\sqrt{914}}{7}$

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**Answer Key**

(As per Official NTA Key released on 2 Feb)

**Q1** (1)**Q2** (3)**Q3** (8)**Q4** (1)**Q5** (4)**Q6** (1)**Q7** (2)**Q8** (3)**Q9** (2)**Q10** (1)**Q11** (1)**Q12** (3)**Q13** (1)**Q14** (2)**Q15** (3)**Q16** (36)**Q17** (3)**Q18** (3)**Q19** (3)**Q20** (3501)**Q21** (11)**Q22** (1)**Q23** (1)

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Q1 (1)

$$\vec{u} = (1, -1, -2), \vec{v} = (2, 1, -1), \vec{v} \cdot \vec{w} = 2$$

$$\vec{v} \times \vec{w} = \vec{u} + \lambda \vec{v} \dots\dots\dots(1)$$

Taking dot with  $\vec{w}$  in (1)

$$\vec{w} \cdot (\vec{v} \times \vec{w}) = \vec{u} \cdot \vec{w} + \lambda \vec{v} \cdot \vec{w}$$

$$\Rightarrow 0 = \vec{u} \cdot \vec{w} + 2\lambda$$

Taking dot with  $\vec{v}$  in (1)

$$\vec{v} \cdot (\vec{v} \times \vec{w}) = \vec{u} \cdot \vec{v} + \lambda \vec{v} \cdot \vec{v}$$

$$\Rightarrow 0 = (2 - 1 + 2) + \lambda(6)$$

$$\lambda = -\frac{1}{2}$$

$$\Rightarrow \vec{u} \cdot \vec{w} = -2\lambda = 1$$

Q2 (3)

$$\text{Let } \vec{\beta}_1 = \lambda \vec{\alpha}$$

$$\text{Now } \vec{\beta}_2 = \vec{\beta} - \vec{\beta}_1$$

$$= (\hat{i} + 2\hat{j} - 4\hat{k}) - \lambda(4\hat{i} + 3\hat{j} + 5\hat{k})$$

$$= (1 - 4\lambda)\hat{i} + (2 - 3\lambda)\hat{j} - (5\lambda + 4)\hat{k}$$

$$\vec{\beta}_2 \cdot \vec{\alpha} = 0$$

$$\Rightarrow 4(1 - 4\lambda) + 3(2 - 3\lambda) - 5(5\lambda + 4) = 0$$

$$\Rightarrow 4 - 16\lambda + 6 - 9\lambda - 25\lambda - 20 = 0$$

$$\Rightarrow 50\lambda = -10$$

$$\Rightarrow \lambda = \frac{-1}{5}$$

$$\vec{\beta}_2 = \left(1 + \frac{4}{5}\right)\hat{i} + \left(2 + \frac{3}{5}\right)\hat{j} - (-1 + 4)\hat{k}$$

$$\vec{\beta}_2 = \frac{9}{5}\hat{i} + \frac{13}{5}\hat{j} - 3\hat{k}$$

$$5\vec{\beta}_2 = 9\hat{i} + 13\hat{j} - 15\hat{k}$$

$$5\vec{\beta}_2 \cdot (\hat{i} + \hat{j} + \hat{k}) = 9 + 13 - 15 = 7$$

Q3 (8)

$$\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \quad \vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}, \quad \vec{a} \cdot \vec{c} = 7$$

$$\vec{a} \times \vec{c} - \vec{b} \times \vec{c} = \vec{0},$$

$$(\vec{a} - \vec{b}) \times \vec{c} = \vec{0} \Rightarrow (\vec{a} - \vec{b}) \text{ is paralleled to } \vec{c}$$

$$\vec{a} - \vec{b} = \mu \vec{c}, \text{ where } \mu \text{ is a scalar}$$

$$-2\hat{i} + 7\hat{j} + 2\lambda\hat{k} = \mu \cdot \vec{c}$$

$$\text{Now } \vec{a} \cdot \vec{c} = 7 \text{ gives } 2\lambda^2 + 12 = 7\mu$$

$$\text{And } \vec{b} \cdot \vec{c} = -\frac{43}{2} \text{ gives } 4\lambda^2 + 82 = 43\mu$$

$$\mu = 2 \text{ and } \lambda^2 = 1$$

$$|\vec{a} \cdot \vec{b}| = 8$$

Q4 (1)

$$\vec{b} = \lambda \vec{a} \times (\vec{a} \times \hat{j})$$

$$\Rightarrow \vec{b} = \lambda(-2\hat{i} - 2\hat{j} + 2\hat{k})$$

$$|\vec{b}| = |\vec{a}| \therefore \sqrt{6} = \sqrt{12}|\lambda| \Rightarrow \lambda = \pm \frac{1}{\sqrt{2}}$$

$$\left( \lambda = \frac{1}{\sqrt{2}} \text{ rejected } \because \vec{b} \text{ makes acute angle with y axis} \right)$$

$$\vec{b} = -\sqrt{2}(-\hat{i} - \hat{j} + \hat{k})$$

$$\frac{(3\vec{a} + \sqrt{2}\vec{b}) \cdot \vec{c}}{|\vec{c}|} = 3\sqrt{2}$$

Q5 (4)

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \frac{\vec{b} - \vec{c}}{2}$$

$$\vec{a} \cdot \vec{c} = \frac{1}{2}, \vec{a} \cdot \vec{b} = \frac{1}{2}$$

$$\therefore \vec{b} \cdot \vec{d} = \frac{1}{2}$$

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \vec{a} \cdot (\vec{b} \times (\vec{c} \times \vec{d}))$$

$$= \vec{a} \cdot ((\vec{b} \cdot \vec{d})\vec{c} - (\vec{b} \cdot \vec{c})\vec{d})$$

$$= (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) = \frac{1}{4}$$

Q6 (1)

Let A : (3, -4, 2)

C : (-2, -1, 3)

B : (1, 2, -1)

D : (5, -2α, 4)

A, B, C, D are coplanar points, then

$$\Rightarrow \begin{vmatrix} 1-3 & 2+4 & -1-2 \\ -2-3 & -1+4 & 3-2 \\ 5-3 & -2\alpha+4 & 4-2 \end{vmatrix} = 0$$

$$\Rightarrow \alpha = \frac{73}{17}$$

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Q7 (2)

$$\vec{a} \times \vec{b} = (\hat{i} - \hat{j})$$

Taking cross product with  $\vec{a}$

$$\Rightarrow \vec{a} \times (\vec{a} \times \vec{b}) = \vec{a} \times (\hat{i} - \hat{j})$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} = \hat{i} + \hat{j} + 2\hat{k}$$

$$\Rightarrow \vec{a} - 3\vec{b} = \hat{i} + \hat{j} + 2\hat{k}$$

$$\Rightarrow 2\vec{a} - 6\vec{b} = 2\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\Rightarrow \vec{a} - 6\vec{b} = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

Q8 (3)

$$\begin{vmatrix} \lambda & \mu & 4 \\ -2 & 4 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 0$$

$$\lambda(10) = \mu(2) + 4(-14) = 0$$

$$10\lambda - 2\mu = 56$$

$$5\lambda - \mu = 28 \quad \dots(1)$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \sqrt{54}$$

$$\frac{-2\lambda + 4\mu - 8}{\sqrt{24}} = \sqrt{54}$$

$$-2\lambda + 4\mu - 8 = \sqrt{54 \times 24} \quad \dots(2)$$

By solving equation (1) & (2)

$$\Rightarrow \lambda + \mu = 24$$

**Q9 (2)**

$$\overline{AB} = (\lambda - 1)\vec{a} - 2\vec{b} + 3\vec{c}$$

$$\overline{AC} = 2\vec{a} + 3\vec{b} - 4\vec{c}$$

$$\overline{AD} = \vec{a} - 3\vec{b} + 5\vec{c}$$

$$\begin{vmatrix} \lambda - 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 1)(15 - 12) + 2(-10 + 4) + 3(6 - 3) = 0$$

**Q10 (1)**

$$\vec{a} \times \vec{b} = 15\hat{i} - 20\hat{j} - 25\hat{k}$$

$$\text{Let } \vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\Rightarrow 15x - 20y - 25z + 25 = 0$$

$$\Rightarrow 3x - 4y - 5z = -5$$

$$\text{Also } x + y + z = 4$$

$$\text{and } \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|} = 1 \Rightarrow 4x + 3y = 5$$

$$\Rightarrow \vec{c} = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$\text{Projection of } \vec{c} \text{ or } \vec{b} = \frac{25}{5\sqrt{2}} = \frac{5}{\sqrt{2}}$$

Q11 (1)

$$\vec{r} \times \vec{b} - \vec{c} \times \vec{b} = 0$$

$$\Rightarrow (\vec{r} - \vec{c}) \times \vec{b} = 0$$

$$\Rightarrow \vec{r} - \vec{c} = \lambda \vec{b}$$

$$\Rightarrow \vec{r} = \vec{c} + \lambda \vec{b}$$

And given that  $\vec{r} \cdot \vec{a} = 0$

$$\Rightarrow (\vec{c} + \lambda \vec{b}) \cdot \vec{a} = 0$$

$$\Rightarrow \vec{c} \cdot \vec{a} + \lambda \vec{b} \cdot \vec{a} = 0$$

$$\Rightarrow \lambda = \frac{-\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}}$$

Now  $\vec{r} \cdot \vec{c} = (\vec{c} + \lambda \vec{b}) \cdot \vec{c}$

$$= \left( \vec{c} - \frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}} \vec{b} \right) \cdot \vec{c}$$

$$= |\vec{c}|^2 - \left( \frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}} \right) (\vec{b} \cdot \vec{c})$$

$$= 74 - \left[ \frac{15}{3} \right] 8$$

$$= 74 - 40 = 34$$

Q12 (3)

$$\hat{n} \perp \vec{c} \quad \vec{a} = \alpha \vec{b} - \vec{n}$$

$$\vec{b} \cdot \vec{c} = 12$$

$$\vec{a} \cdot \vec{c} = \alpha(\vec{b} \cdot \vec{c}) - \vec{n} \cdot \vec{c}$$

$$\vec{a} \cdot \vec{c} = \alpha(\vec{b} \cdot \vec{c})$$

$$|\vec{c} \times (\vec{a} \times \vec{b})| = |(\vec{c} \cdot \vec{b})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b}|$$

$$= |(\vec{c} \cdot \vec{b})\vec{a} - \alpha(\vec{b} \cdot \vec{c})\vec{b}|$$

$$= |(\vec{c} \cdot \vec{b})| |\vec{a} - \alpha \vec{b}|$$

$$= 12 \times (|\vec{n}|)$$

$$= 12 \times 1$$

$$= 12$$

Q13 (1)

$$\vec{a} = \lambda \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{b} = \hat{i} - \lambda \hat{j} + 2\hat{k}$$

$$\Rightarrow (\vec{b} - \vec{a}) \times ((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})) = 8\hat{i} - 40\hat{j} - 24\hat{k}$$

$$\Rightarrow ((\vec{a} - \vec{b}) \cdot (\vec{a} + \vec{b}))(\vec{a} \times \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$$

$$\Rightarrow 8(\vec{a} \times \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$$

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \lambda & 2 & -3 \\ 1 & -\lambda & 2 \end{vmatrix}$$

$$= (4 - 3\lambda)\hat{i} - (2\lambda + 3)\hat{j} + (-\lambda^2 - 2)\hat{k}$$

$$\Rightarrow \lambda = 1$$

$$\therefore \vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\Rightarrow \vec{a} + \vec{b} = 2\hat{i} + \hat{j} - \hat{k}, \vec{a} - \vec{b} = 3\hat{j} - 5\hat{k}$$

$$\Rightarrow (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 0 & 3 & -5 \end{vmatrix} = 2\hat{i} + 10\hat{j} + 6\hat{k}$$

$$\therefore \text{required answer} = 4 + 100 + 36 = 140$$

**Q14 (2)**

$$\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$$

$$\vec{b} \cdot \vec{c} = \vec{b} \cdot (2\vec{a} \times \vec{b}) - 3\vec{b} \cdot \vec{b}$$

$$= -3|\vec{b}|^2$$

$$= -48$$

Q15 (3)

$$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a} + \vec{b} - \vec{c}|^2$$

$$2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a} = 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} - 2\vec{c} \cdot \vec{a}$$

$$4\vec{a} \cdot \vec{c} = 0$$

B is incorrect

$$|\vec{a} + \lambda \vec{c}|^2 \geq |\vec{a}|^2$$

$$\lambda^2 c^2 \geq 0$$

True  $\forall \lambda \in \mathbb{R}$  (A) is correct.

Q16 (36)

$$|\vec{a}| = \sqrt{14}, |\vec{b}| = \sqrt{6} \quad |\vec{a} \times \vec{b}| = \sqrt{48}$$

$$|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2 \times |\vec{b}|^2$$

$$\Rightarrow (\vec{a} \cdot \vec{b})^2 = 84 - 48 = 36$$

Q17 (3)

$$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$$

$$(\vec{r} - \vec{c}) \times \vec{b} = 0, \quad \vec{r} \cdot \vec{a} = 0$$

$$\Rightarrow \vec{r} - \vec{c} = \lambda \vec{b}$$

$$\text{Also, } (\vec{c} + \lambda \vec{b}) \cdot \vec{a} = 0$$

$$\Rightarrow \vec{a} \cdot \vec{c} + \lambda (\vec{a} \cdot \vec{b}) = 0$$

$$\therefore \lambda = \frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} = \frac{-8}{5}$$

$$\vec{r} = \frac{5(5\hat{i} - 3\hat{j} + 3\hat{k}) - 8(\hat{i} - \hat{j} + 2\hat{k})}{5}$$

$$\vec{r} = \frac{17\hat{i} - 7\hat{j} + \hat{k}}{5}$$

$$|\vec{r}|^2 = \frac{1}{25}(289 + 50)$$

$$25|\vec{r}|^2 = 339$$

Q18 (3)

// Equation of Plane:

$$(2\hat{i} + a\hat{j} + 4\hat{k}) \cdot [(x-2)\hat{i} + (y-a)\hat{j} + (z-4)\hat{k}] = 0$$

$$\Rightarrow 2x + ay + 4z = 20 + a^2$$

$$\Rightarrow A \equiv \left( \frac{20 + a^2}{2}, 0, 0 \right)$$

$$B \equiv \left( 0, \frac{20 + a^2}{a}, 0 \right)$$

$$C \equiv \left( 0, 0, \frac{20 + a^2}{4} \right)$$

//  $\Rightarrow$  Volume of tetrahedron

$$= \frac{1}{6} [\vec{a} \vec{b} \vec{c}]$$

$$= \frac{1}{6} \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$\Rightarrow \frac{1}{6} \left( \frac{20 + a^2}{2} \right) \cdot \left( \frac{20 + a^2}{a} \right) \cdot \left( \frac{20 + a^2}{4} \right) = 144$$

$$\Rightarrow (20 + a^2)^3 = 144 \times 48 \times a$$

$$\Rightarrow a = 2$$

$$\Rightarrow \text{Equation of plane is } 2x + 2y + 4z = 24$$

$$\text{Or } x + y + 2z = 12$$

$$\Rightarrow (3, 0, 4) \quad \text{Not lies on the Plane}$$

$$x + y + 2z = 12$$

Q19 (3)

$$\vec{b} \cdot 3\vec{c} \times \vec{a}$$

$$\vec{a} \times (2\vec{b} + 3\vec{c}) = 0$$

$$\vec{a} = \lambda(2\vec{b} + 3\vec{c})$$

$$|\vec{a}|^2 = \lambda^2 |2\vec{b} + 3\vec{c}|^2$$

$$|\vec{a}|^2 = \lambda^2 (4|\vec{b}|^2 + 9|\vec{c}|^2 + 12\vec{b} \cdot \vec{c})$$

$$31 = 31\lambda^2 \Rightarrow \lambda = \pm 1$$

$$\vec{a} = \pm(2\vec{b} + 3\vec{c})$$

$$\frac{|\vec{a} \times \vec{c}|}{|\vec{a} \cdot \vec{b}|} = \frac{2|\vec{b} \times \vec{c}|}{2\vec{b} \cdot \vec{b} + 3\vec{c} \cdot \vec{b}}$$

$$|\vec{b} \times \vec{c}|^2 = |\vec{b}|^2 |\vec{c}|^2 - (\vec{b} \cdot \vec{c})^2 = \frac{3}{4}$$

$$\frac{|\vec{a} \times \vec{c}|}{|\vec{a} \cdot \vec{b}|} = \frac{2 \times \frac{\sqrt{3}}{2}}{2 \cdot 1 + 3} = \frac{\sqrt{3}}{5}$$

#MathBoleTohMathonGo

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$$[\vec{u} \vec{v} \vec{w}] = \vec{u} \cdot (\vec{v} \times \vec{w})$$

$$\min. (|\vec{u}| |\vec{v} \times \vec{w}| \cos \theta) = -\alpha \sqrt{3401}$$

$$\Rightarrow \cos \theta = -1$$

$$|\vec{u}| = \alpha \text{ (Given)}$$

$$|\vec{v} \times \vec{w}| = \sqrt{3401}$$

$$\vec{v} \times \vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 2 & -3 \\ 2\alpha & 1 & -1 \end{vmatrix}$$

$$\vec{v} \times \vec{w} = \hat{i} - 5\alpha \hat{j} - 3\alpha \hat{k}$$

$$|\vec{v} \times \vec{w}| = \sqrt{1 + 25\alpha^2 + 9\alpha^2} = \sqrt{3401}$$

$$34\alpha^2 = 3400$$

$$\alpha^2 = 100$$

$$\alpha = 10 \quad (\text{as } \alpha > 0)$$

$$\text{So } \vec{u} = \lambda (\hat{i} - 5\alpha \hat{j} - 3\alpha \hat{k})$$

$$|\vec{u}| = \sqrt{\lambda^2 + 25\alpha^2 \lambda^2 + 9\alpha^2 \lambda^2}$$

$$\alpha^2 = \lambda^2 (1 + 25\alpha^2 + 9\alpha^2)$$

$$100 = \lambda^2 (1 + 34 \times 100)$$

Q21 (11)

$$A(2, 6, 2) \quad B(-4, 0, \lambda), C(2, 3, -1) \quad D(4, 5, 0)$$

$$\text{Area} = \frac{1}{2} |\overrightarrow{BD} \times \overrightarrow{AC}| = 18$$

$$\overrightarrow{AC} \times \overrightarrow{BD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -3 \\ 8 & 5 & -\lambda \end{vmatrix}$$

$$= (3\lambda + 15)\hat{i} - \hat{j}(-24) + \hat{k}(-24)$$

$$\overrightarrow{AC} \times \overrightarrow{BD} = (3\lambda + 15)\hat{i} + 24\hat{j} - 24\hat{k}$$

$$= \sqrt{(3\lambda + 15)^2 + (24)^2 + (24)^2} = 36$$

$$= \lambda^2 + 10\lambda + 9 = 0$$

$$= \lambda = -1, -9$$

$$|\lambda| \leq 5 \Rightarrow \lambda = -1$$

$$5 - 6\lambda = 5 - 6(-1) = 11$$

Q22 (1)

$$\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$$

$$\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$$

Q23 (1)

$$\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}$$

$$\vec{b} = \hat{i} + \hat{k}$$

$$\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{r} \times \vec{a} = \vec{c} \times \vec{a} \Rightarrow (\vec{r} - \vec{c}) \times \vec{a} = 0$$

$$\therefore \vec{r} = \vec{c} + \lambda \vec{a}$$

$$\vec{r} \cdot \vec{b} = 0 \Rightarrow \vec{c} \cdot \vec{b} + \lambda \vec{b} \cdot \vec{a} = 0$$

$$-2 + \lambda(7) = 0 \Rightarrow \lambda = \frac{2}{7}$$

$$\therefore \vec{r} = \vec{c} + \frac{2\vec{a}}{7} = \frac{1}{7}(11\hat{i} - 11\hat{k})$$